A Practical MRI Grading System for Lumbar Foraminal Stenosis

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Inventory Category:	Clinical grading / reference standard
Comparability / Priority:	Very High | Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Clinical grading standard | Sagittal lumbar MRI
Dataset & Cohort:	nan (Sample: nan)
Disease / Target:	Neural foraminal stenosis
Model / AI Method:	Manual Lee grading
Evaluation Metrics:	Interobserver agreement
Key Finding / Relevance:	nan
Thesis Context (Ch 2):	The RSNA task contains left/right neural foraminal narrowing; this is a core clinical precursor.

Abstract / Summary

ObjectiveThis study aimed to evaluate the reproducibility of a new grading system for lumbar foraminal stenosis.Materials and methodsFour grades were developed for lumbar foraminal stenosis on the basis of sagittal MRI. Grade 0 refers to the absence of foraminal stenosis; grade 1 refers to mild foraminal stenosis showing perineural fat obliteration in the two opposing directions, vertical or transverse; grade 2 refers to moderate foraminal stenosis showing perineural fat obliteration in the four directions without morphologic change, both vertical and transverse directions; and grade 3 refers to severe foraminal stenosis showing nerve root collapse or morphologic change. A total of 576 foramina in 96 patients were analyzed (from L3-L4 to L5-S1). Two experienced radiologists independently assessed the sagittal MR images. Interobserver agreement between the two radiologists and intraobserver agreement by one reader were analyzed using kappa statistics.ResultsAccording to reader 1, grade 1 foraminal stenosis was found in 33 foramina, grade 2 in six, and grade 3 in seven. According to reader 2, grade 1 foraminal stenosis was found in 32 foramina, grade 2 in six, and grade 3 in eight. Interobserver agreement in the grading of foraminal stenosis between the two readers was found to be nearly perfect (kappa value: right L3-L4, 1.0; left L3-L4, 0.905; right L4-L5, 0.929; left L4-L5, 0.942; right L5-S1, 0.919; and left L5-S1, 0.909). In intraobserver agreement by reader 1, grade 1 foraminal stenosis was found in 34 foramina, grade 2 in eight, and grade 3 in seven. Intraobserver agreement in the grading of foraminal stenosis was also found to be nearly perfect (kappa value: right L3-L4, 0.883; left L3-L4, 1.00; right L4-L5, 0.957; left L4-L5, 0.885; right L5-S1, 0.800; and left L5-S1, 0.905).ConclusionThe new grading system for foraminal stenosis of the lumbar spine showed nearly perfect interobserver and intraobserver agreement and would be helpful for clinical study and routine practice.